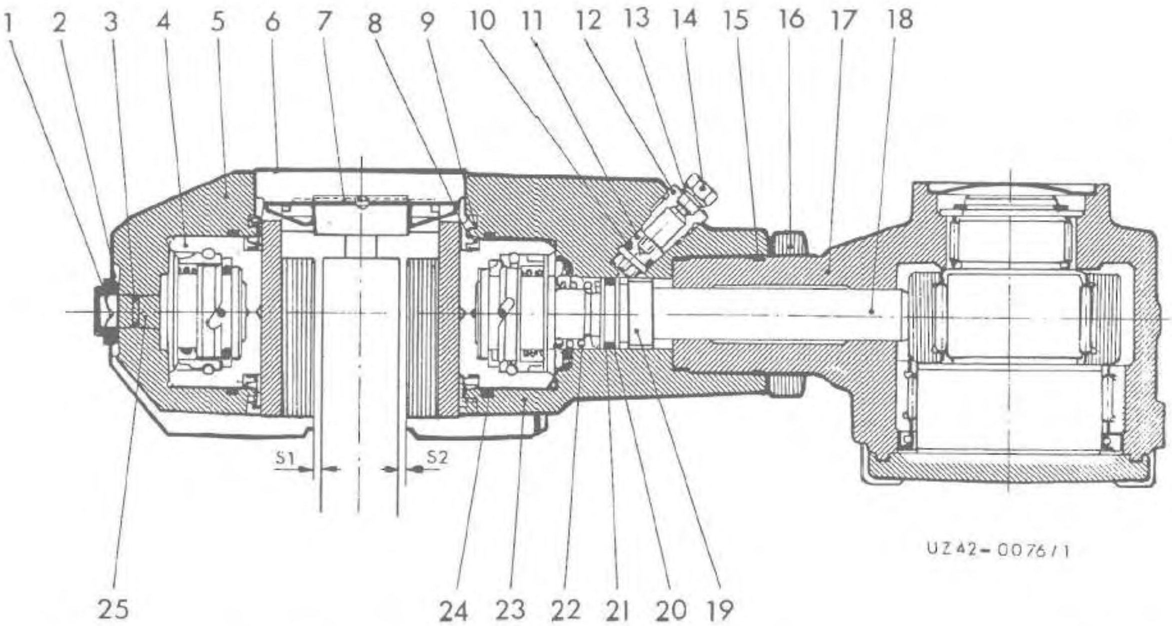
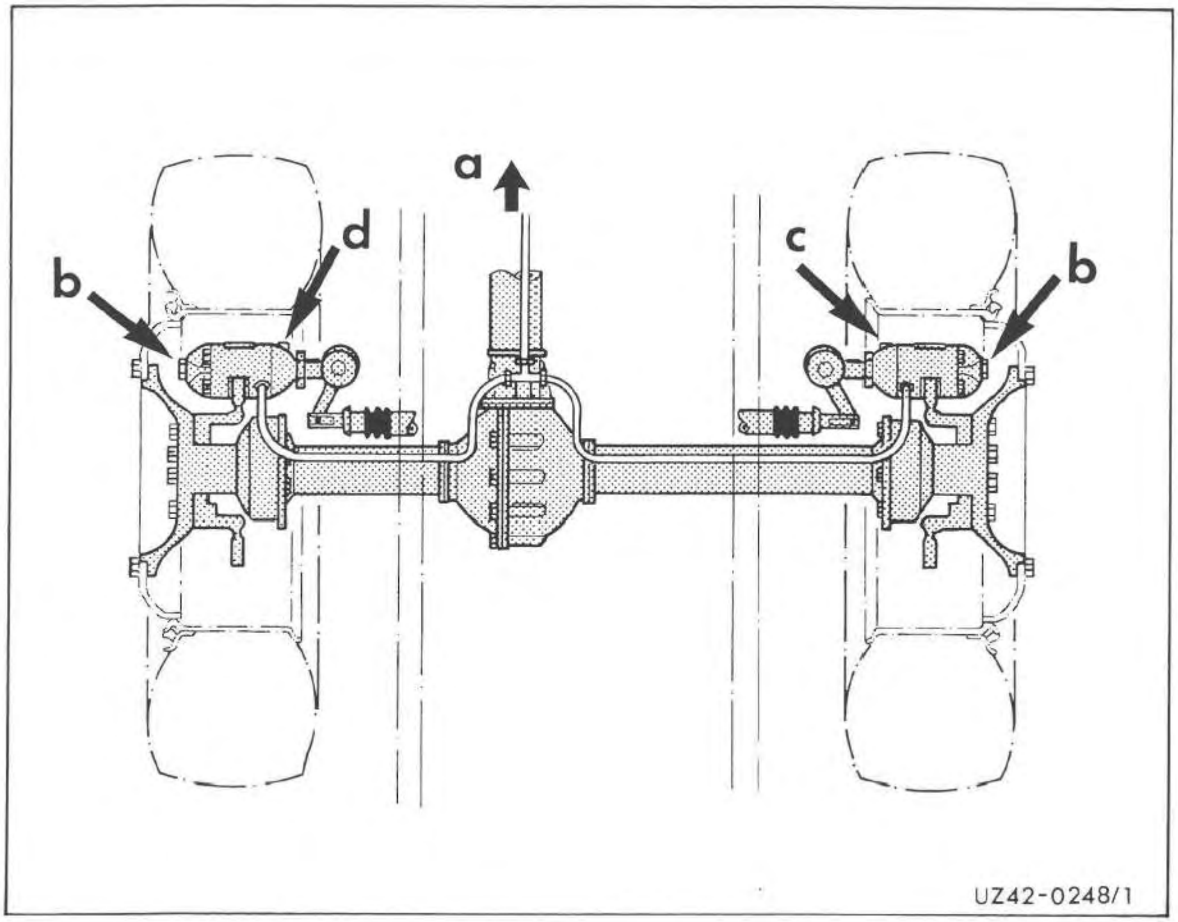
Sectional View



Fixed Caliper

1	Hexagon nut (self-locking)	14	Screw plug
2	Lock washer	15	Sealing ring
3	Sealing ring	16	Hexagon nut
4	Piston	17	Actuating lever
5	Housing cover	18	Pin
6	Cover plate	19	Drive spindel
7	Expander spring	20	Support ring
8	Shield	21	Sealing ring
9	Dust cap	22	Pressure spring
10	Sealing ring	23	Housing flange
11	Restoring shaft	24	Sealing ring
12	Threaded sleeve	25	Adjustment spindel
13	Sealing ring		

Directions of Rotation Specifications for Adjusting the Fixed Combined Caliper



a = direction of travel

	b	c	d
Piston to brake disk	Conterclockwise	Clockwise	Clockwise
Piston from brake disk	Clockwise	Counterclockwise	Counterclockwise

Troubleshooting

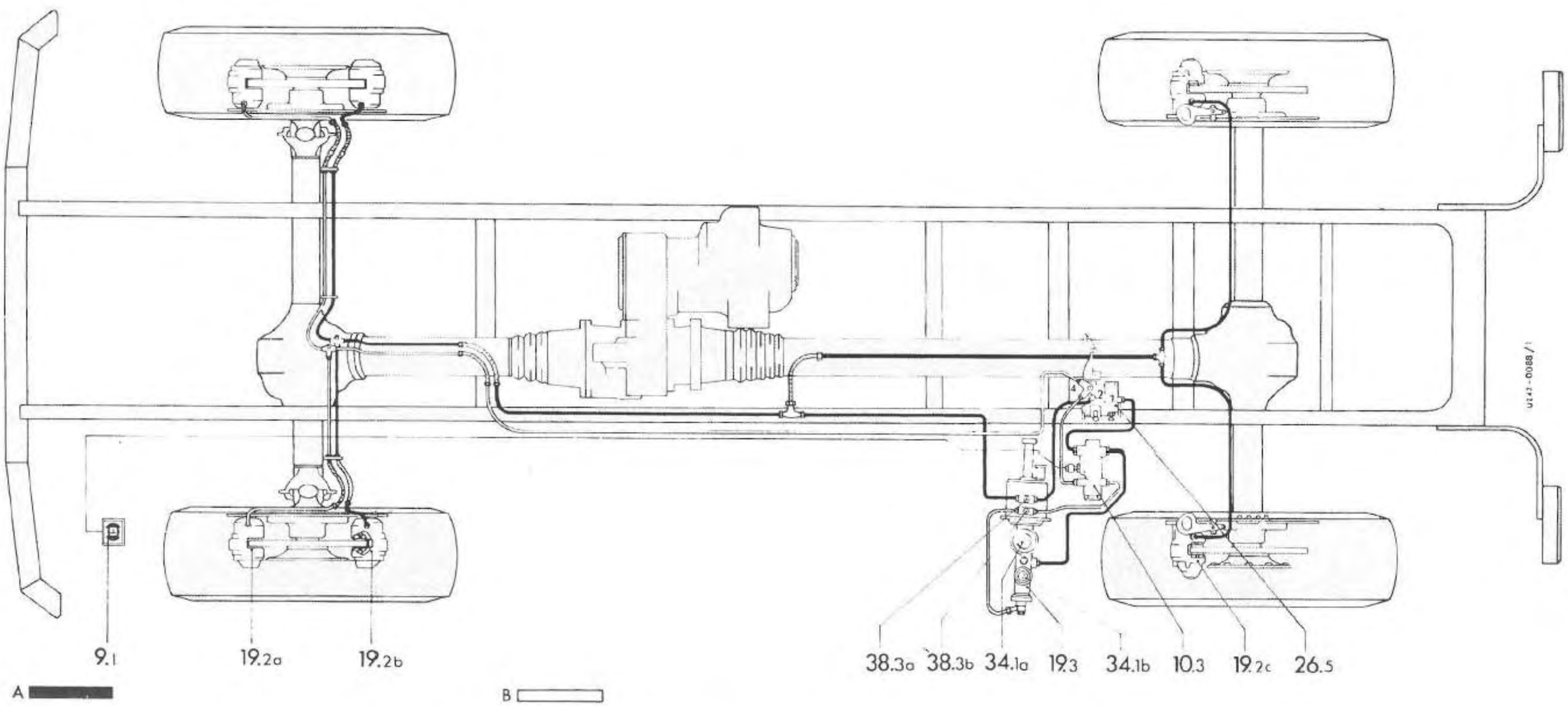
Hydraulic Brake System

Fault	Cause	Remedy
No or inadequate braking action.	Air in the system	Bleed system, top up brake fluid.
	Spring or linkage to ALB modulator (west) broken or detached.	Attach spring, linkage or renew. Adjust ALB according to type plate specifications.
The effect of the brake fades during partial braking.	Leakages in lines or damaged or unsuitable sealing rings in master cylinder or brake caliper.	The lines must be sealed. Renew damaged sealing rings.
Warning lamp for brake system lights up in the instrument cluster during driving.	Tandem master brake cylinder faulty. Hydraulic brake circuit I or II failed.	Check, if necessary, recondition or renew. Find leaks in brake system and eliminate.
IMPORTANT! Stop immediately.	Worn brake linings.	Renew brake pads.
	Brake fluid level too low in the reservoir or empty.	Fill reservoir with specified brake fluid up to the upper marking. Check hydraulic brake system, seal and bleed if necessary.
Brake is applied by itself or heats up during driving.	Equalization bores blocked in tandem master brake cylinder.	Clean equalization bores.
	No play at brake pedal.	Adjust pedal play to 10 – 20 mm.
	Brake valve, cable jammed or stiff causing residual pressure in preload cylinder.	Release brake valve, cable, exchange if necessary.
Brake overheats at one axle.	Hydraulic hose jammed or kinked. Valve effect in hose.	Renew hydraulic hose.
Air repeatedly found in the hydraulic brake system.	Breather hole in cap of reservoir clogged. Reservoir or connecting line from the reservoir to the master cylinder leaking at connections.	Clean breather hole in cap. Tighten reservoir or connecting line. Bleed brake system.
	Air is sucked into the system via the brake pressure regulator.	Check brake pressure regulator, renew if necessary.

Fault	Cause	Remedy
No braking action and warning lamp for brake system lights up during driving. **CAUTION!** Stop immediately	Air in brake system, brake system leaking.	Briefly release brake pedal and quickly fully depress twice in which case resistance must be felt. If the brakes have no effect, stop the vehicle by applying the handbrake.
	No brake fluid in reservoir.	Determine leakage in brake system and repair. Bleed brake system.
No braking action	Seized brake pads	Clean soiled wells in the brake caliper and brake disk. Check dust cap, renew if necessary. Check spring clip and retaining pins, renew if necessary. Install specified brake pads in sets on one axle. **Note:** Adjust clearance at combined brake caliper.
	Seized piston in brake caliper.	Remove corrosion in the brake caliper cylinders and piston. Release piston. Renew gaskets.
Insufficient braking action or uneven braking action.	Worn brake pads	Install specified brake pads in sets on one axle.
	Wet or soiled brakes	Dry by driving while slightly applying brake, clean if necessary. Check brake disk cover.
	Incorrect brake pads installed.	Install specified brake pads in sets on one axle.
	Brake disk surface is scored.	Surface grind or renew brake disks. Observe tolerance specifications.
	Unequal tyre pressure.	Adjust tyre pressure.
	Brake pads oil-stained.	Seal wheel hubs or steering knuckle.
	Uneven pad wear	Install specified brake pads in sets on one axle. Clean soiled wells in brake caliper and brake disk. Remove corrosion in caliper cylinders and piston, release piston. Check piston return (clearance), release piston if necessary.

Fault	Cause	Remedy
	Pressure applied by ALB too low during full braking.	Check correct actuating pressure for ALB in accordance with type plate specifications and adjust if necessary.
	Closing pressure of pressure reducing valve too low. For this reason, the full brake pressure is not obtained during full braking.	Correct closing pressure of pressure regulating valves.
Brake becomes too hot	Brake pads do not release from the brake disks.	Clean soiled wells in the brake caliper and brake disk. Check dust cap, renew if necessary.
		Renew spring clip with retaining pins.
		Check wheel hub mounting, secure if necessary.
		Check brake disk cover.
		Remove corrosion in the brake caliper cylinders and piston, ensure piston moves smoothly.
	Piston return does not operate correctly in brake caliper.	Check piston return setting (clearance), release piston if necessary. Renew seals.
	Brake valve does not allow air to escape completely.	Check operation of brake valve. Renew in the case of internal stiffness. Check cable and ensure it moves smoothly. Check or adjust pedal play.
	Spring-loaded brake does not release. Release pressure too low.	Check hand brake valve, renew if necessary.
Noisy brake (Squeeking)	Incorrect brake pads installed.	Install specified brake pads in sets on one axle.
	Brake pad vitrified. Brake disks scored.	Surface grind the pad surface and brake disks, renew if necessary.
		Clean soiled wells in brake caliper and brake disk. Check dust cap, renew if necessary.
	20° position of piston collar not correct in brake caliper.	Check 20° position of piston collar in brake caliper, correct if necessary.
		Renew spring clip with retaining pins.

Brake Diagram 435.110/111/113

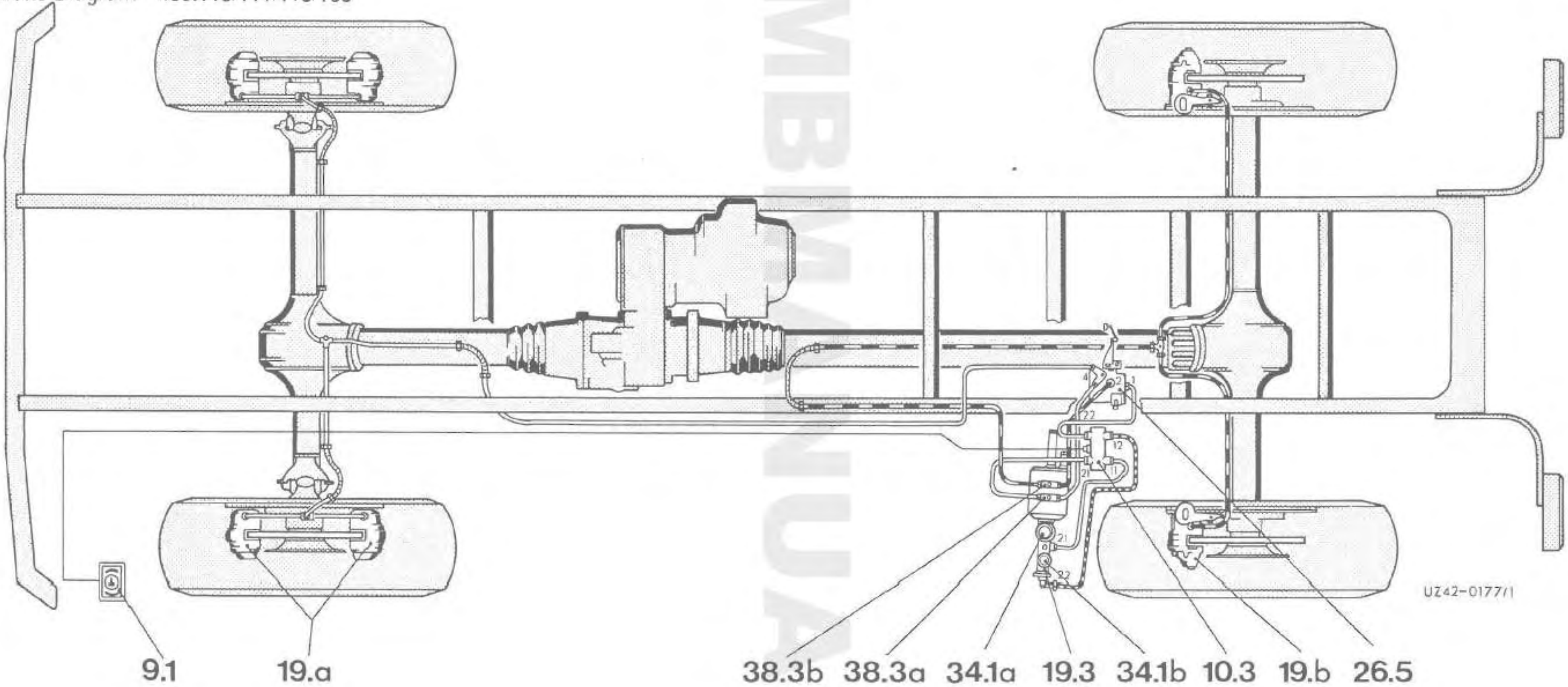


Dual-circuit hydraulic brake system with ALB on rear axle and ½ front axle
to chassis No. 435.110/111 – 086 854, 435.113 – 084 641

A	Circuit **1** modulated, rear axle, ALB modulator and ½ front axle (rear fixed caliper)
B	Circuit 2 unmodulated, front axle (front fixed caliper)
9.1	Warning lamp pressure difference (SA 35 975)
10.3	Electrical pressure difference warning switch (SA 35 975)
19.2a	Fixed caliper, circuit 2 unmodulated
19.2b	Fixed caliper, circuit 1 modulated
19.2c	Fixed combined caliper, circuit 1 modulated
19.3	Master brake cylinder/dual-circuit
26.5	Tandem modulator ALB
34.1a	Hydraulic tank circuit 1
34.1b	Hydraulic tank circuit 2
38.3a	Test connection front axle at front, circuit 2 unmodulated
38.3b	Test connection rear axle, circuit 1 modulated, ½ front axle rear.

See 43.11 — 1.5/1 for 1.5/28 for symbols.

Brake Diagram 435.110/111/113/160



A B C

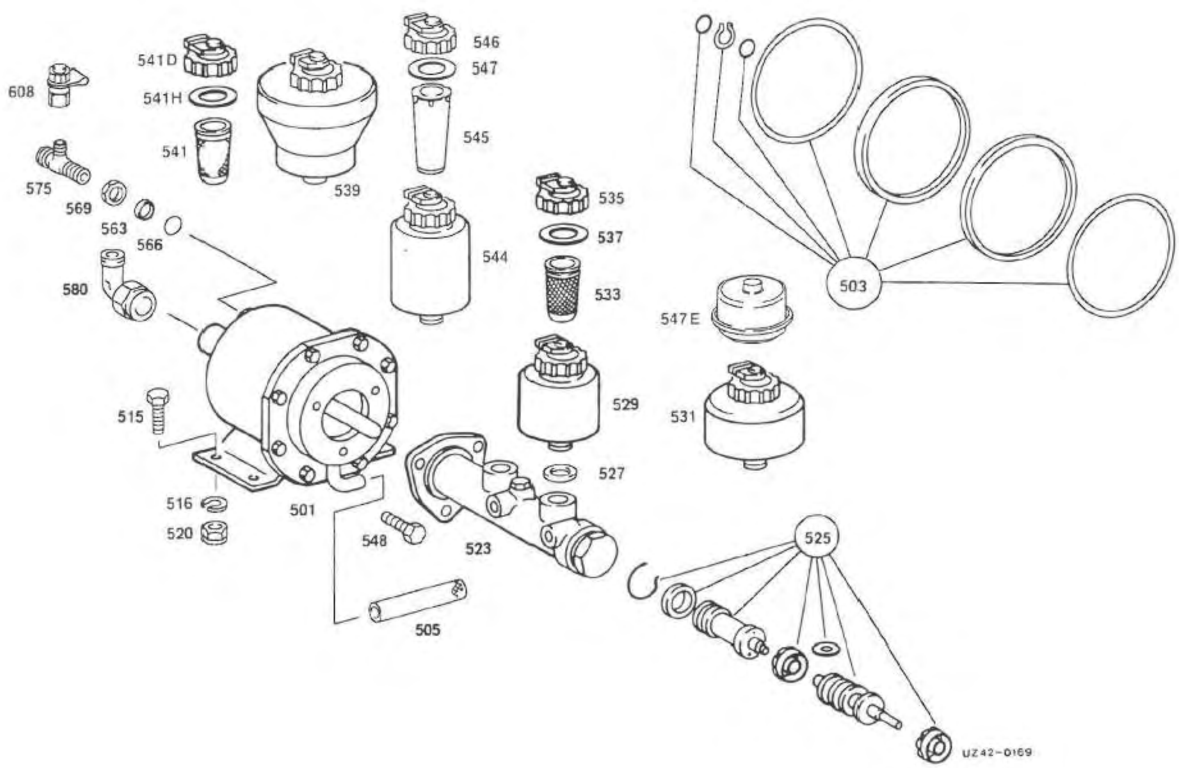
Dual-circuit hydraulic brake system with ALB on rear axle
from chassis No. 435.110/111 – 086 855, 435.113 – 084 642

A	Front axle	(circuit 1)
B	Rear axle unmodulated before ALB	
C	Rear axle modulated after ALB	(circuit 2)

9.1 Pressure difference warning lamp (SA 35 975)
10.3 Pressure difference warning switch (SA 35 975)
19.a Fixed caliper unmodulated circuit 1
19.b Fixed combined caliper modulated circuit 2
19.3 Master brake cylinder dual-circuit
26.5 Single-circuit ALB modulator
34.1a Equalizing tank circuit 1
34.1b Equalizing tank circuit 2
38.3a Test connection brake circuit 1 unmodulated
38.3b Test connection brake circuit 2 modulated

See 43.11 – 1.5/1 to 1.5/28 for symbols.

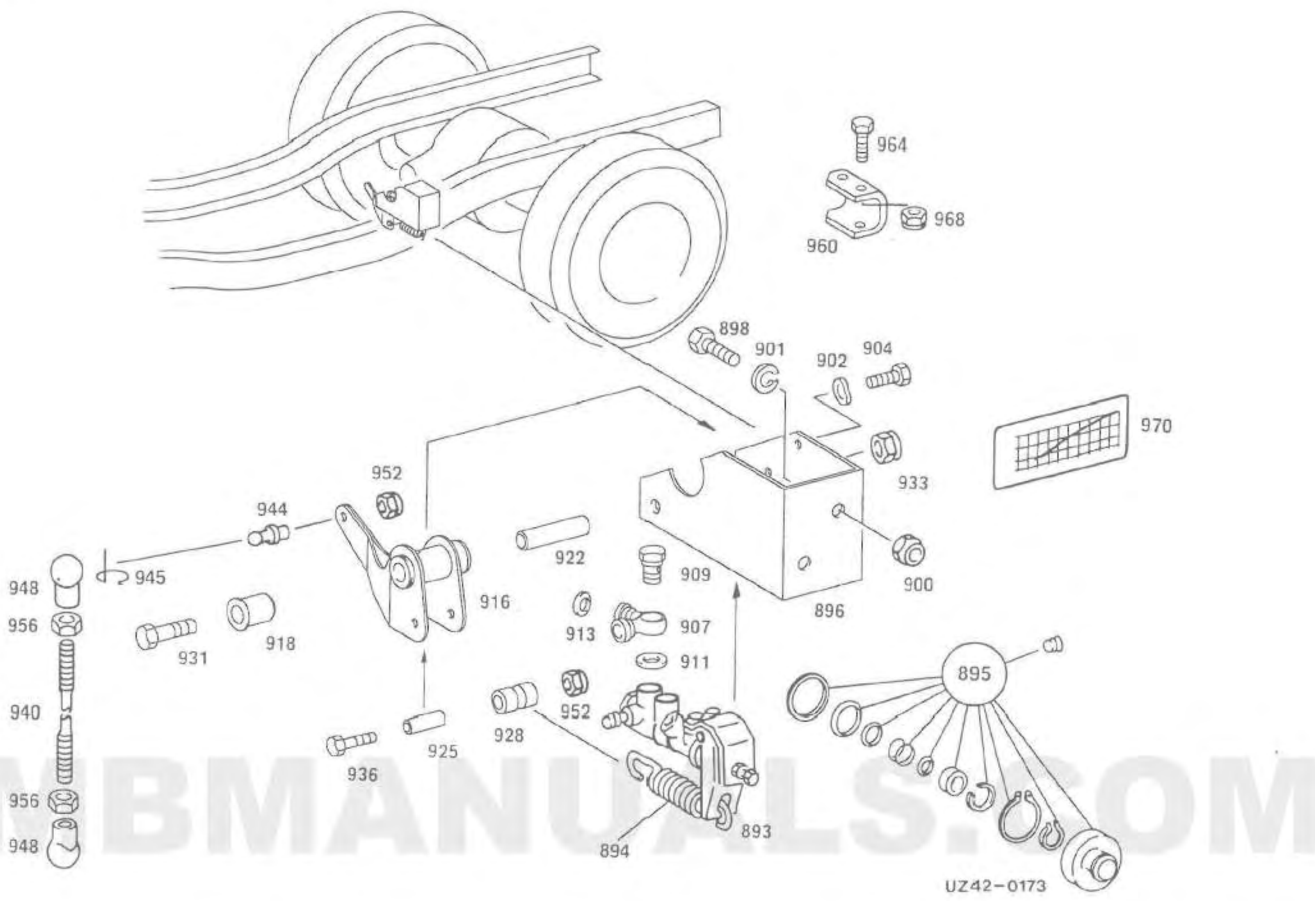
Exploded View



Preload Cylinder, Master Cylinder

501 Preload cylinder
503 Repair kit
505 Hose
515 Bolt
516 Lock washer
520 Nut
523 Tandem master cylinder
525 Repair kit
527 Sealing ring
529 Reservoir 0,13 liter
531 Reservoir 0,24 liter
533 Gauze filter
535 Cap
537 Sealing ring
539 Reservoir 0,5 liter
541 Gauze filter
541D Cap
541H Sealing ring
544 Reservoir 0,33 liter
545 Gauze filter
546 Cap
547 Sealing ring
547E Protective cap
548 Bolt
563 Pressure spring
566 Sealing ring
569 Nut
575 Pipe socket
580 Pipe socket
608 Test connection

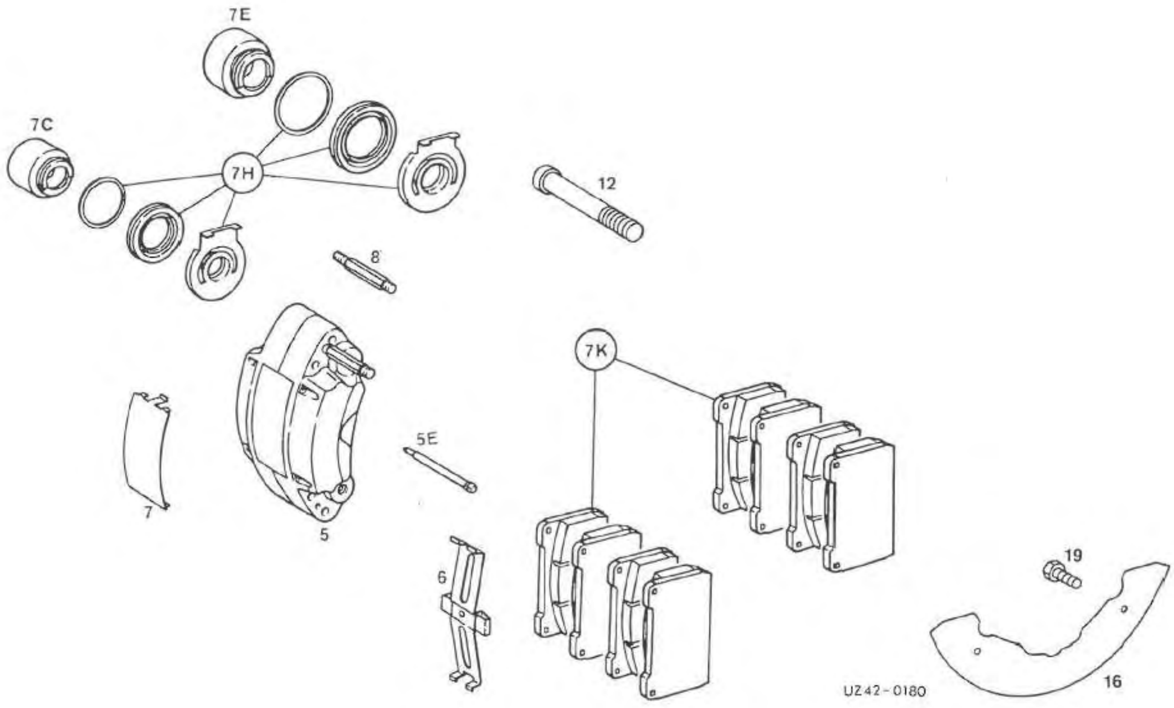
Exploded View



ALB Modulator

893 Brake pressure regulator
894 Spring
895 Repair kit
896 Bracket
898 Bolt
900 Nut
901 Lock washer
902 Spring washer
904 Bolt
907 Distributor
909 Banjo bolt
911 Sealing ring
913 Sealing ring
916 Lever
918 Spacer tube
922 Bushing
925 Bushing
928 Bearing
931 Bolt
933 Nut
936 Bolt
940 Pull rod
944 Ball pin
945 Clip
948 Ball socket
952 Nut
956 Nut
960 Bracket
964 Bolt
968 Nut
970 Information plate (ALB adjustment)

Exploded View

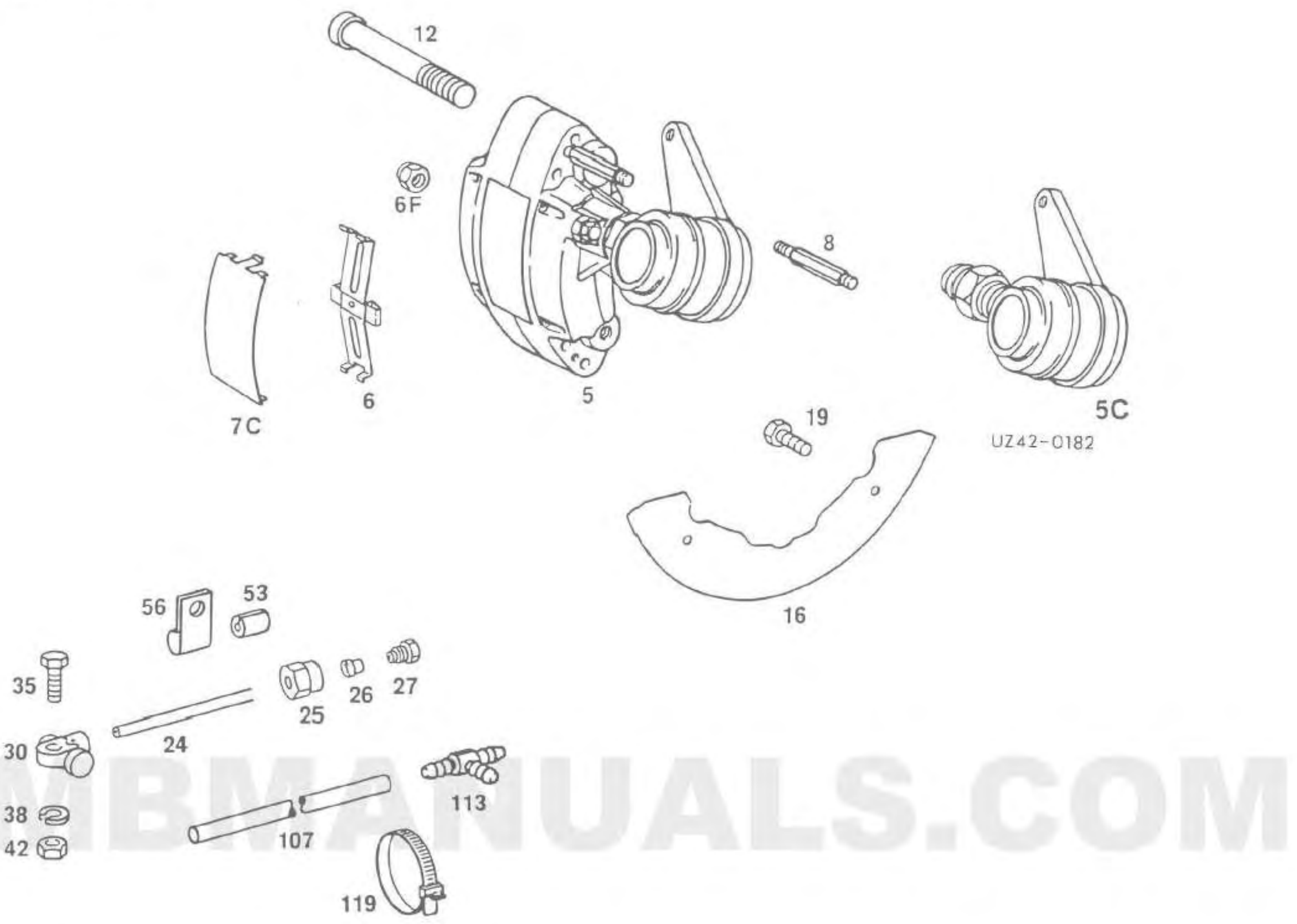


Brake Components, Front Axle

5 Brake caliper
5E Pin
6 Spring
7 Guard plate
7C Piston 48 mm
7E Piston 57 mm
7H Repair kit
7K Repair kit brake pads
8 Bleeder valve
12 Bolt
16 Guard plate
19 Bolt

747.2

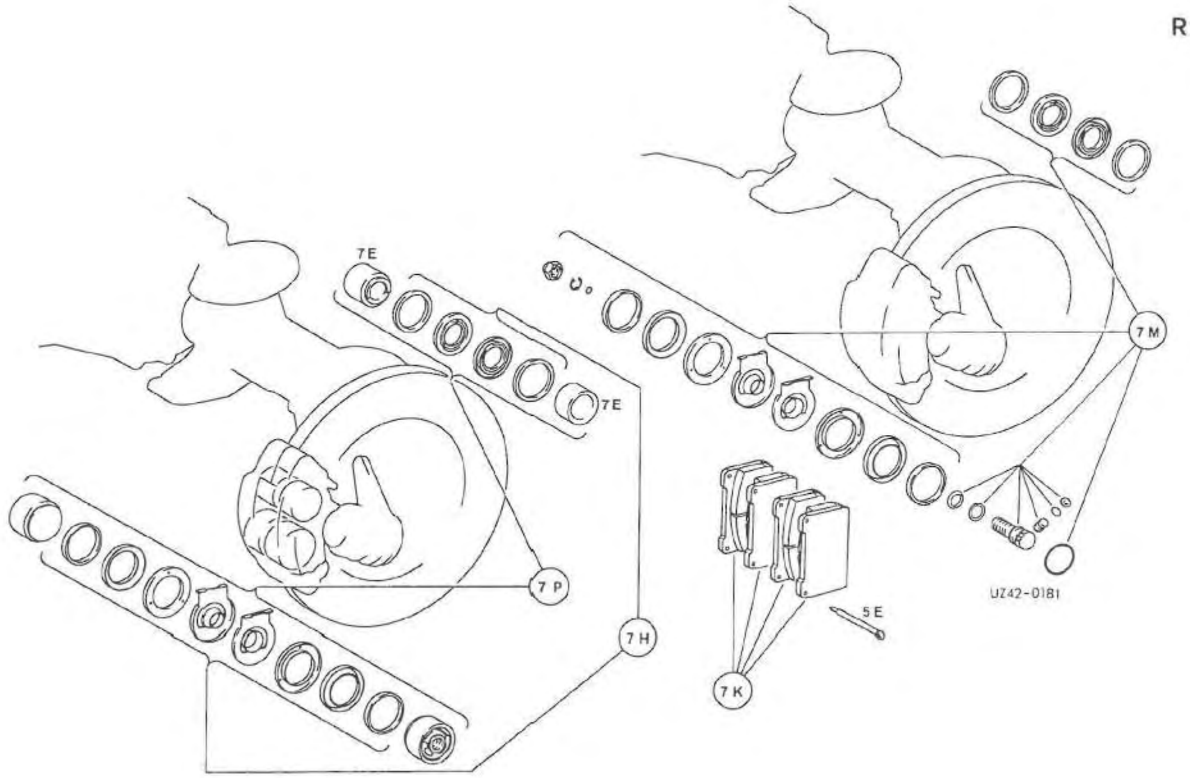
Exploded View



Brake Components, rear Axle

5	Brake caliper
5C	Brake lever
6	Spring
6F	Nut
7C	Guard plate
8	Bleeder valve
12	Caliper fastening bolt
16	Guard plate
19	Bolt
24	Line
25	Union nut
26	Bite ring
27	Union screw
30	Distributor
35	Bolt
38	Lock washer
42	Nut
53	Hose
56	Clamp
107	Hose
113	Connection piece
119	Clip

Exploded View



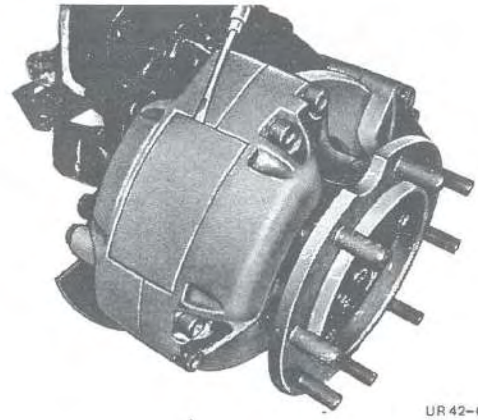
Brake Components, Rear Axle

5E Pin
7E Piston 48 mm
7F Piston 57 mm
7H Repair kit
7K Repair kit brake linings
7M Repair kit with adjustment
7P Repair kit with piston

Removal

1 Remove wheel.

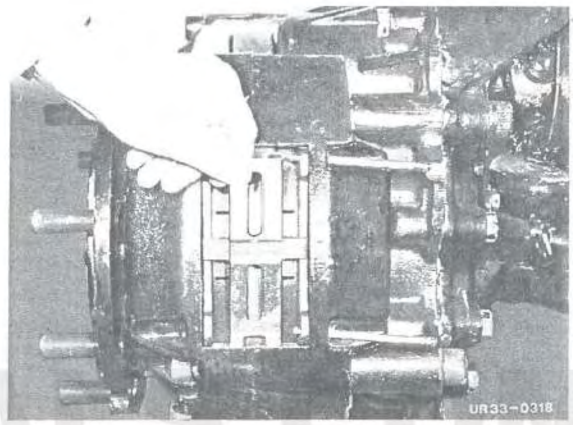
2 Remove cover plate,
va'id for brake lining wear indicator.



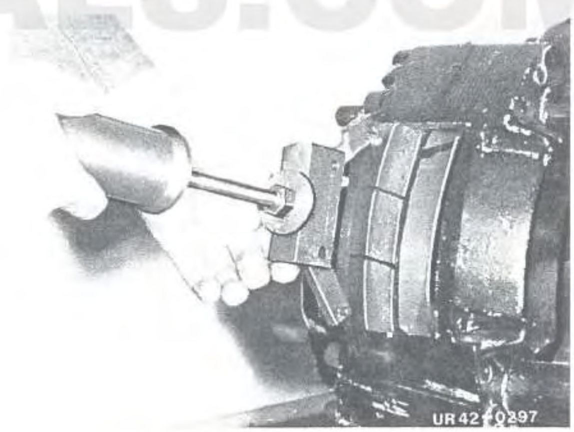
UR 42-0677

3 Drive out retaining pin.

4 Remove spring clip.

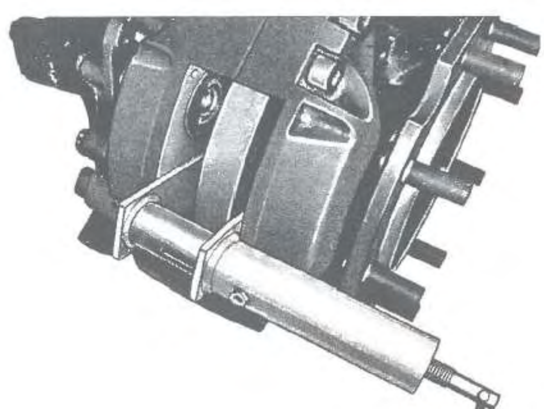


5 Remove brake pads using special tool No. 8. If brake pads are difficult to remove, use special tool No. 7.

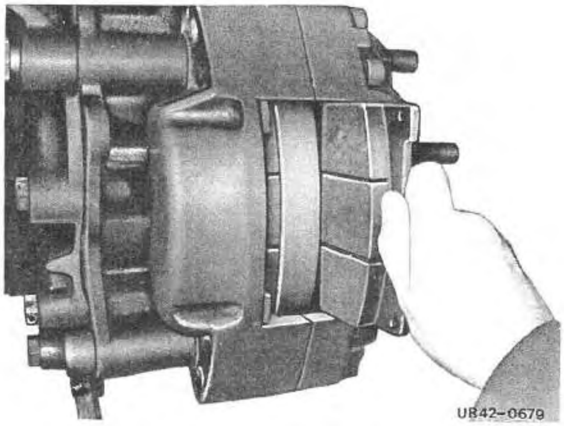


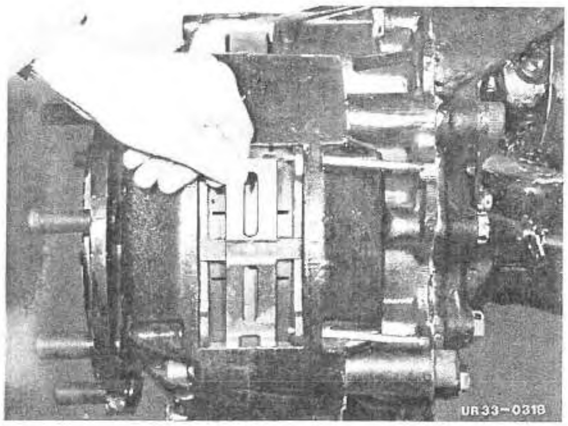
6 Using special tool No. 11, move piston in initial position.

7 Clean caliper shaft with cleaning agent No. 1.



UR 42-0678





Installation

1 Install brake pads.

Note: Only exchange brake pads in complete sets on one axle.

IMPORTANT!
The thicknesses of the backplates of the brake linings differ for the front and rear axle. Front axle 6 mm, rear axle 8 mm.

2 Fit spring clip.

3 Drive in retaining pin.

4 Fit guard plate.
Valid for brake lining wear indicator.

5 Fit wheel and tighten wheel nuts.
Tightening torques, see 1.3/1.

Note: Retighten wheel nuts after 2 hours of operation or 50 km.

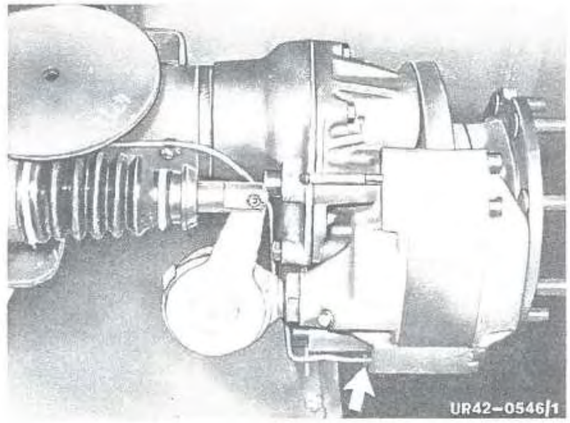
6 Actuate service brake several times and check that the wheel moves freely.

7 Check brake fluid level and top up if necessary.

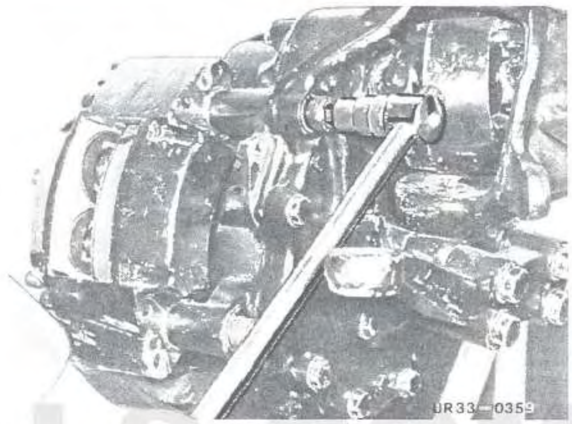
Removal

1 Remove brake pads, see 2.1/1.

2 Disconnect brake line and close off.

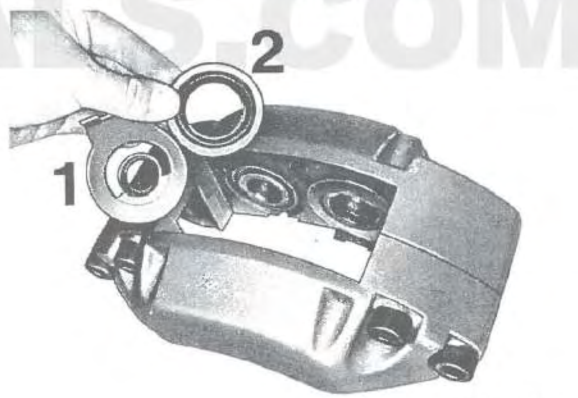


3 Detach fixed caliper using special tool No. 1.



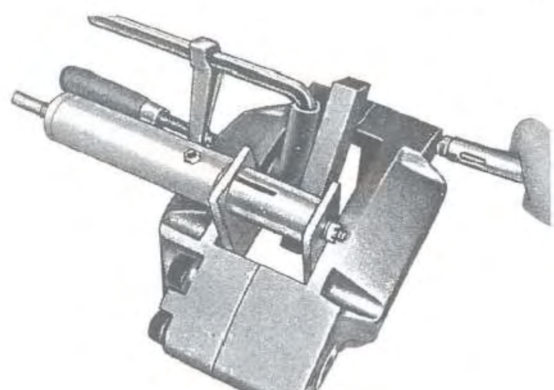
4 Remove guard and dust cap.

1 Guard cap
2 Dust cap

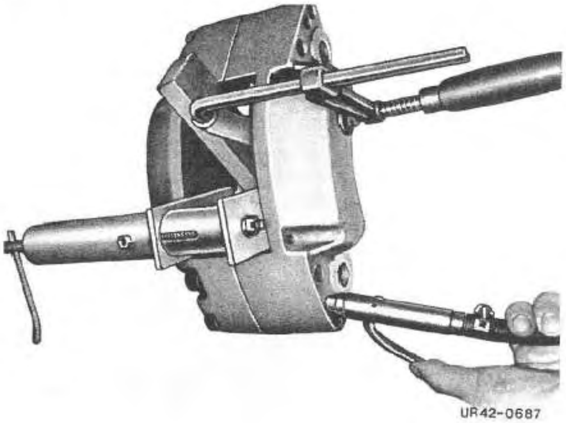


UR 42-0685/1

5 Hold both 57 diameter pistons with special tool No. 11 and piston 48 diameter with screw clamp. Press out the free piston using compressed air.



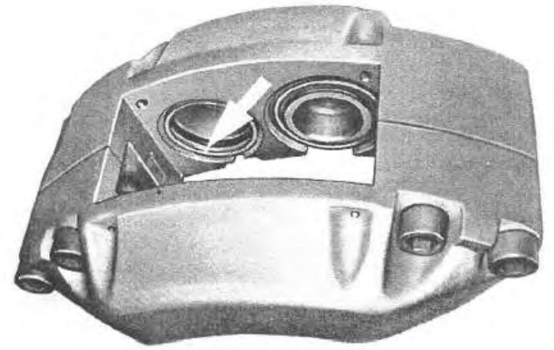
UR 42-0686



UR42-0687

6 Close off the piston bore with special tools No. 11 and No. 4. Press out the second piston 48 mm diameter.

7 Close off piston bore 48 mm diameter using special tool No. 4. Hold one 57 mm diameter piston using a screw clamp. Blow out the other piston.
Remove the last piston in the same way.



UR42-0688/1

8 Remove sealing rings.
Clean caliper and pistons with cleaning agent No. 1.
Check removed parts for wear, renew if necessary.

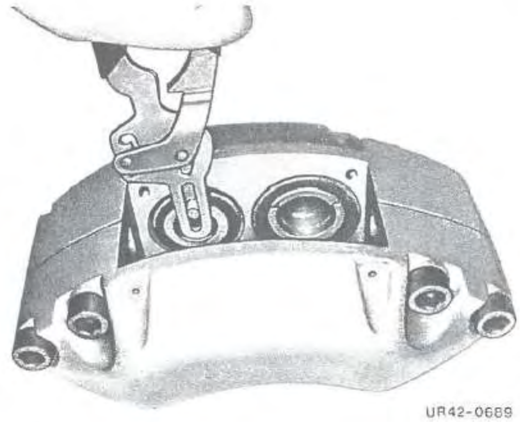
Installation

Note: Use a new set of seals, coat the cylinder bores, pistons, and sealing rings with paste No. 4.

1 Install sealing ring and piston, ensuring the piston is not tilted.

Note: The piston shoulder must face outwards in the direction of the edge of the caliper.
If necessary, set correctly using special tool No. 9.

2 Half fill the dust cap with paste No. 4, insert in groove on the piston and place on to the shoulder in the cylinder bore using special tools No. 11, No. 10 and No. 14 (15). Ensure it is seated correctly.



UR42-0689

3 Install guard using special tools No. 11, Nr. 10 and No. 12 (13).

4 Install fixed caliper using special tool No. 1.

5 Fit brake pads, see 2.1/1.

6 Install spring clip and retaining pin.

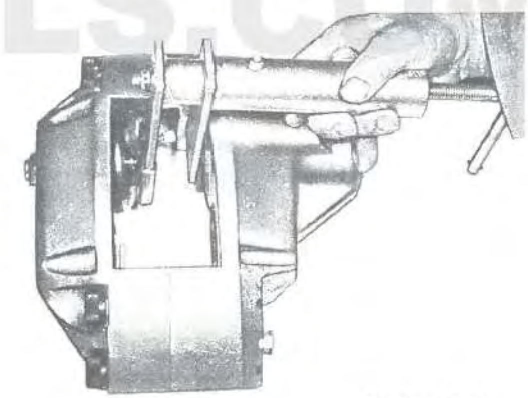
7 Fit cover plate.

8 Connect brake line.

9 Fit wheel.
Thightening torque, see 1.3/1.

Note: Retighten wheel nuts after 50 km.

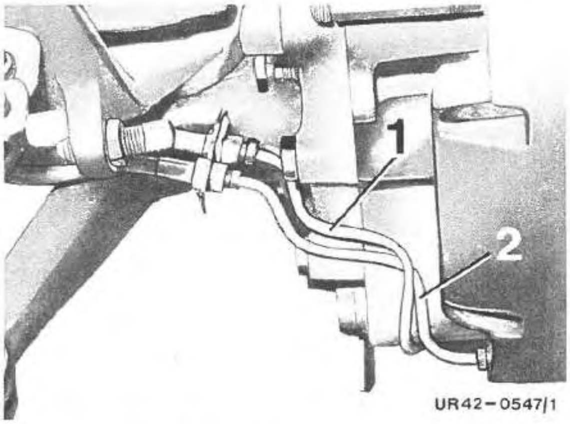
10 Bleed brake system, see 5.1/1.



UR42-0286

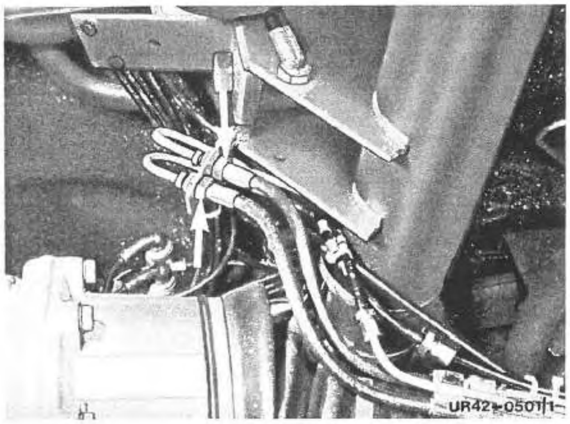
Checking

To Chassis end No.:
435.110/111 – 086 854
435.113 – 084 641

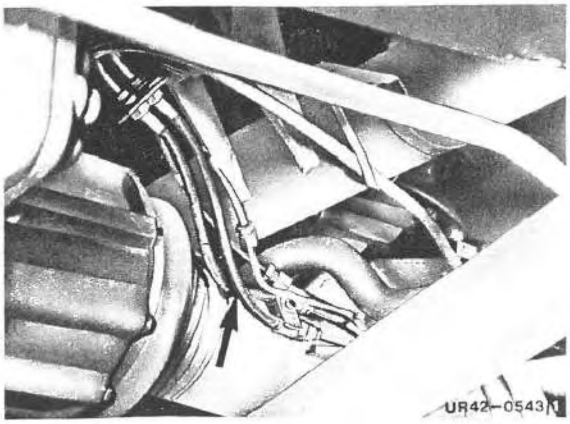


1 Check wether the connections for brake circuits 1 and 2 are connected in accordance with the hydraulic diagram, see 1.8/1.

1 Brake circuit 1
2 Brake circuit 2



2 The rubber hoses must not be crossed at the connecting point on the torque ball casing.



Important!

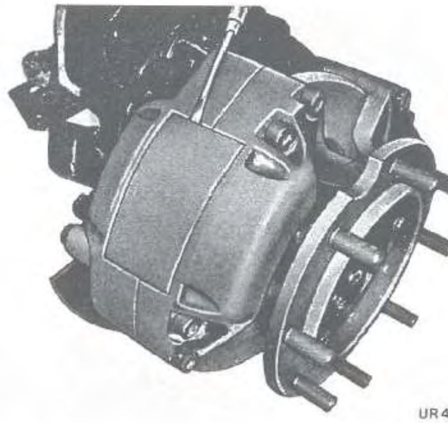
3 The rubber hoses must be installed crossed at the connection point in vehicles with brake lines which are not crossed at the fixed caliper.

Removal

1 Detach wheel.

2 Release parking brake.

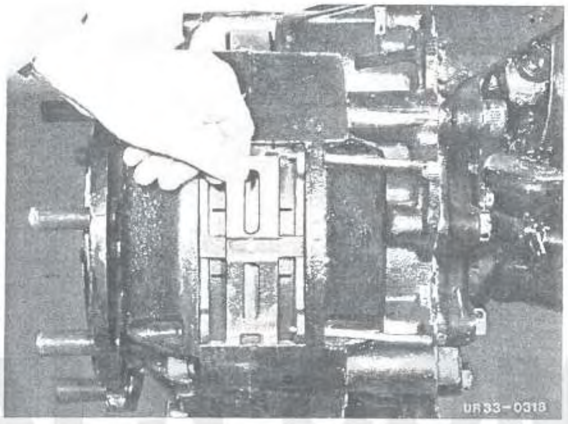
3 Remove guard plate.
Valid for brake lining wear indicator.



UR 42-0677

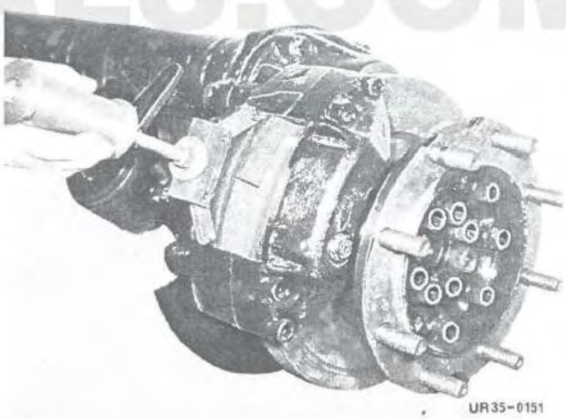
4 Drive out retaining pins.

5 Remove spring clip.



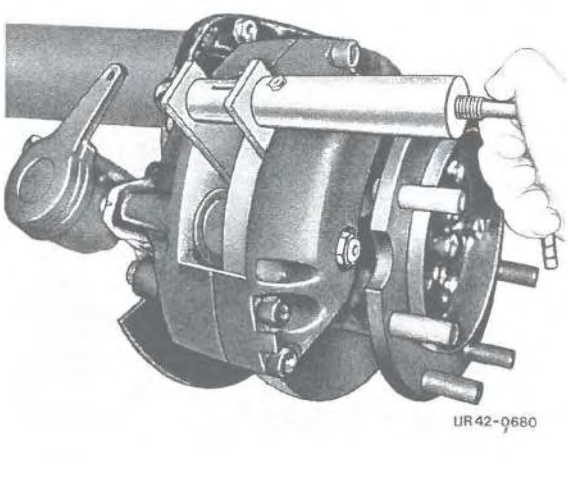
UR 33-0318

6 Withdraw brake pads using special tool No. 8.

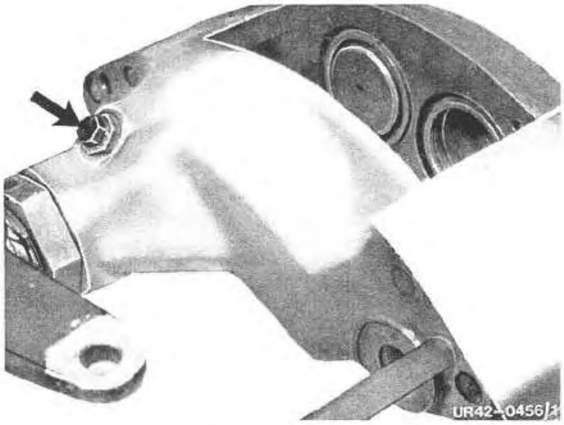


UR 35-0151

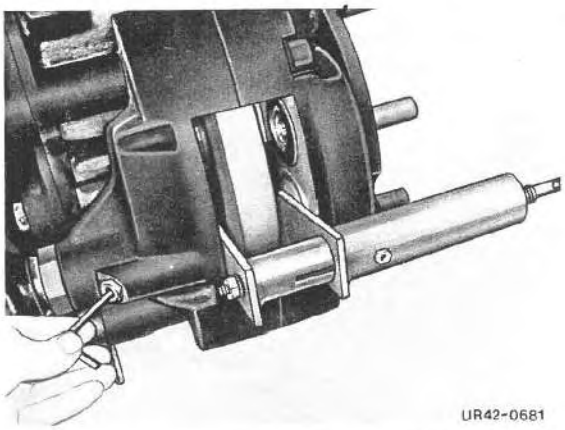
7 Move 48 mm diameter piston to initial position using special tool No. 11.



UR 42-0680

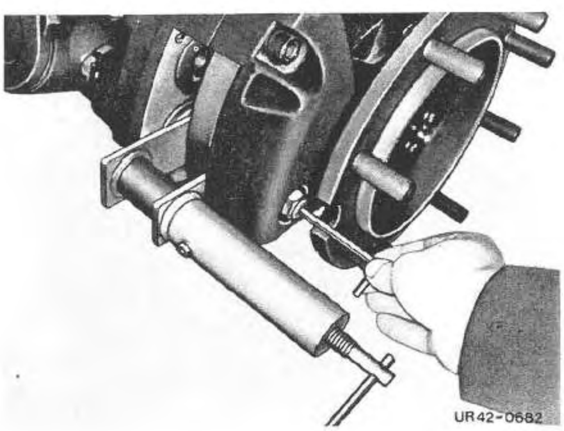


9 Unscrew screw plug together with sealing ring.



10 Move piston on flange side to its initial position by turning the restoring shaft.
Direction of rotation, see 1.5/1.

Note: Maintain pressure applied to piston with special tool No. 11.



11 Loosen counternut and move piston on cover side to initial position by turning the adjusting screw.
Direction of rotation, see 1.5/1.

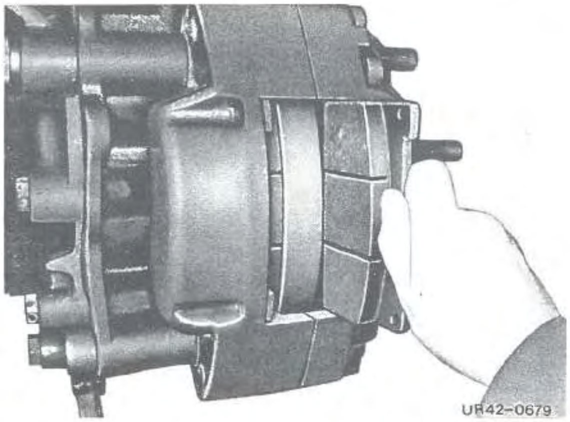
Note: Maintain pressure applied to piston with tool No. 11.

12 Clean caliper shaft with cleaning agent No. 1.

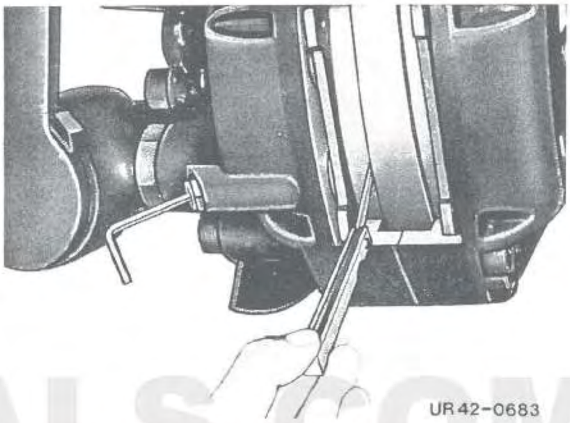
Installation

1 Install brake pads only in sets on each axle.

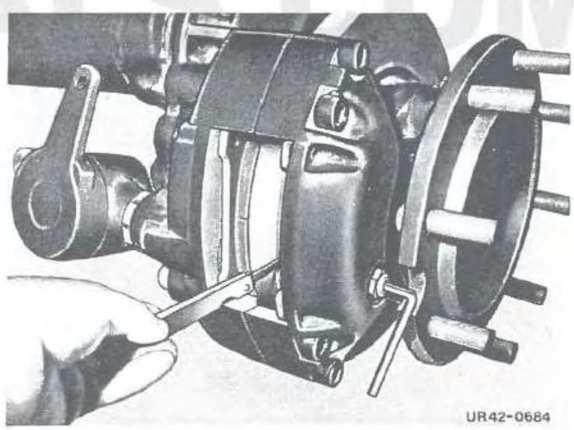
Important!
The thickness of the brake lining backplates differ for the front axle and rear axle. Front axle 6 mm, rear axle 8 mm.



2 Carry out basic brake pad adjustment by checking the clearance on the flange side between the brake pad and brake disk and adjusting if necessary.
Direction of rotation, see 1.5/1.
Permissible clearance, see 1.1/1.
Tighten screw plug together with new copper sealing ring.
Tightening torque, see 1.3/1.



3 Check clearance on cover side using feeler gauge between brake pad and brake disk and adjust if necessary.
Direction of rotation, see 1.5/1.
Permissible clearance, see 1.1/1.
Tighten counternut.
Tightening torque, see 1.3/1.



4 Fit spring clip.

5 Drive in retaining pins.

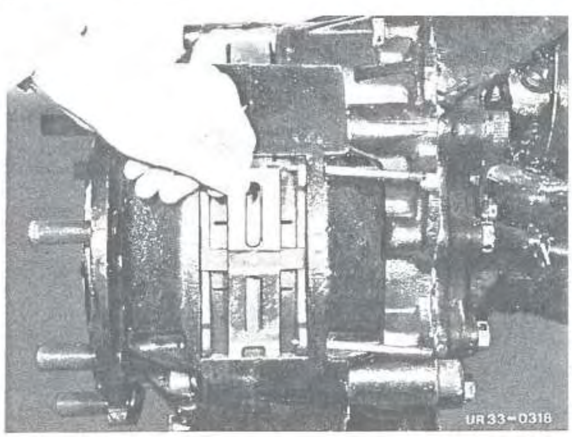
6 Install guard plate for spring caliper with brake lining wear indicator.

7 Attach wheel and tighten wheel nuts.
Tightening torques, see 1.3/1.

Note: Retighten wheel nuts after 50 km.

8 Actuate service brake and parking brake several times and check that wheel moves freely.

9 Check brake fluid level and top up if necessary.



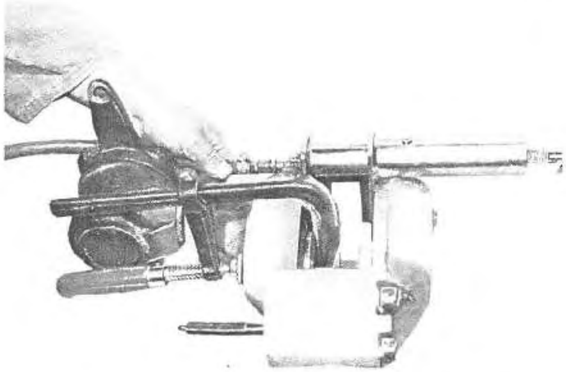
Diassembly

Note:
Fixed calipers should be diassembled only by technicians in authorized workshops.

The item numbers ( ) refer to page 1.4/1.

1 Remove fixed caliper.

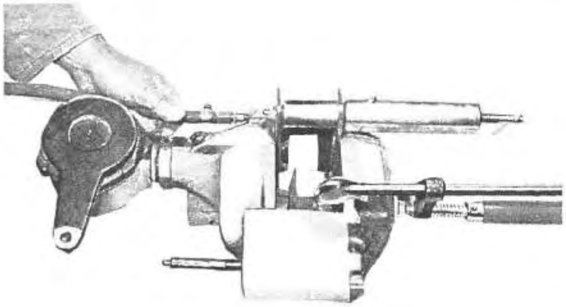
2 Remove guard caps (8) and protective caps (9) from pistons.



UR42–0454

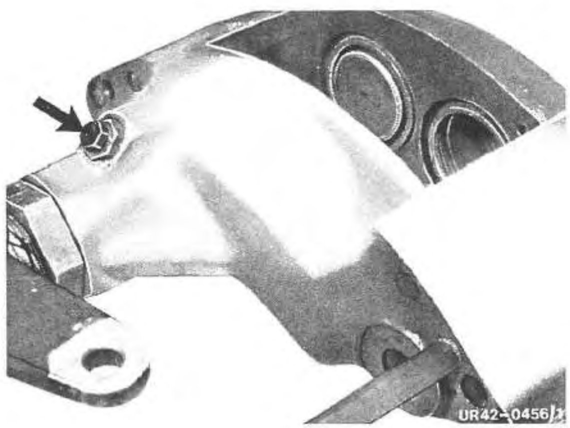
3 Secure both 57 mm diameter pistons (4) with special tool No. 11 and one 48 mm diameter piston with screw clamp, forcing out the free piston with compressed air.

4 Using special tools No. 11 and No. 4, close open piston bore, remove second 48 mm diameter piston as described in step No. 3.



UR42–0455

5 Remove screw plug (14) with sealing ring (13).

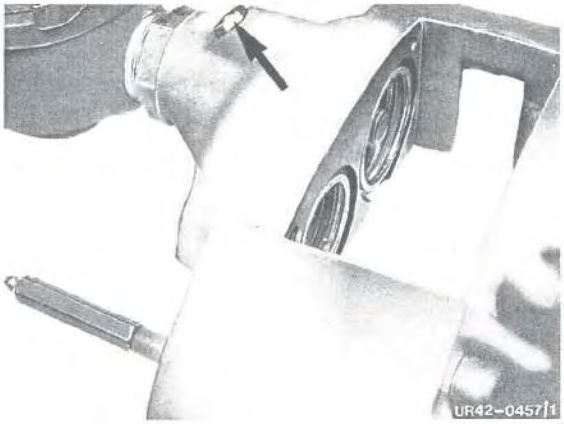


UR42–0456

6 Turn restoring shaft (11) and remove 57 mm diameter piston from flange section (23).
Direction of rotation, see 1.5/1.

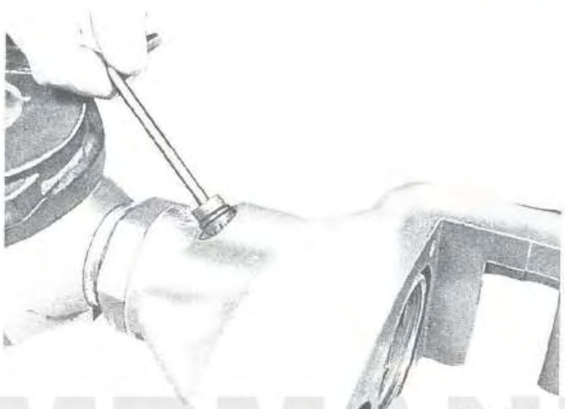
Note: If piston resists removal with restoring shaft (11), force out piston using compressed air as described in step No. 3. To do so, close 48 mm diameter piston bores with sealing plates and special tool No. 11. After removal, force adjusting fixture back into position, centering adjusting fixture in piston bore and carefully pressing into position until securing ring latches distinctly and audibly.

UR42–0470



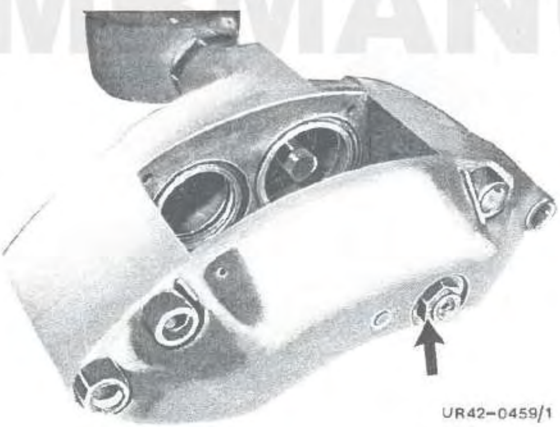
UR42-0457/1

7 Remove threaded sleeve (12).



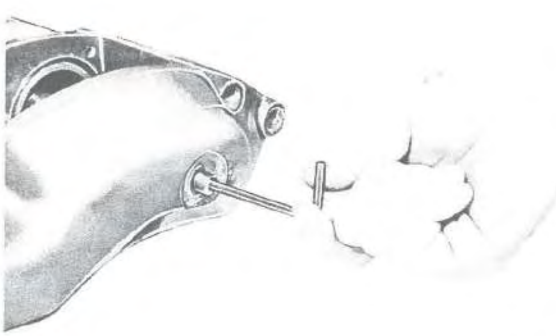
UR42-0458

8 Take out restoring shaft (11).



UR42-0459/1

9 Unscrew hexagon nut (1).

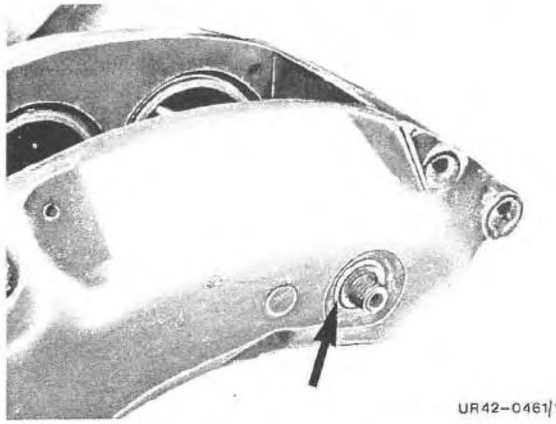


UR42-0460

10 Turn adjusting spindle (25) and withdraw 57 mm diameter piston (4) from cover section (5).
Direction of rotation, see 1.5/1.

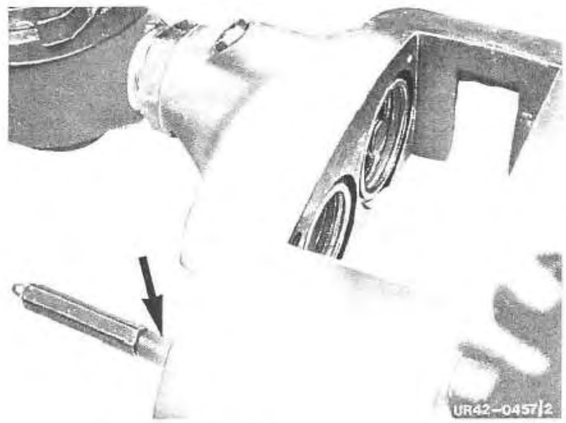
Note: If piston resists removal with adjusting screw (25), remove piston using compressed air as described in steps No. 3 and No. 6.

11 Release adjusting spindle (25) at (2) and remove.



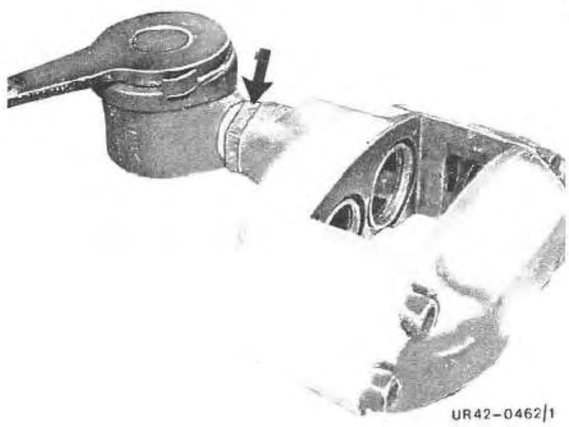
UR42-0461/1

12 Unscrew bleeder valve.



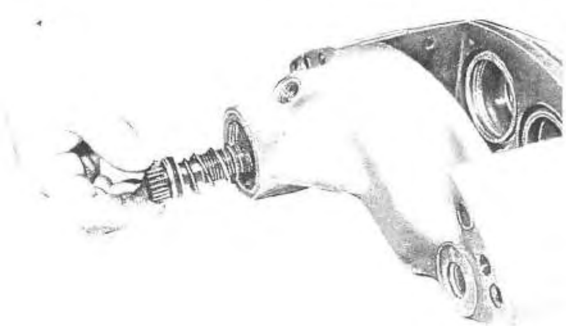
UR42-0457/2

13 Loosen hexagon nut (16) and unscrew actuating lever (17).

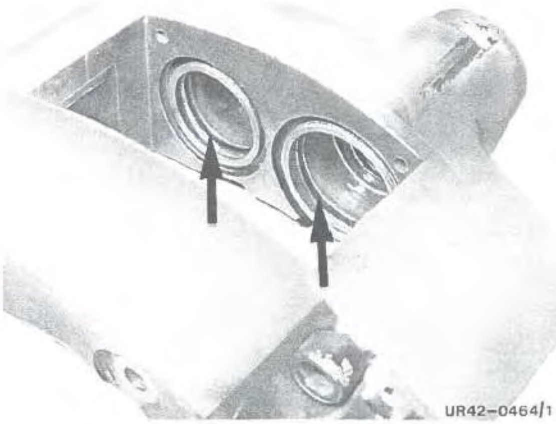


UR42-0462/1

14 Remove drive spindle (19) together with compression spring (22).



UR42-0463

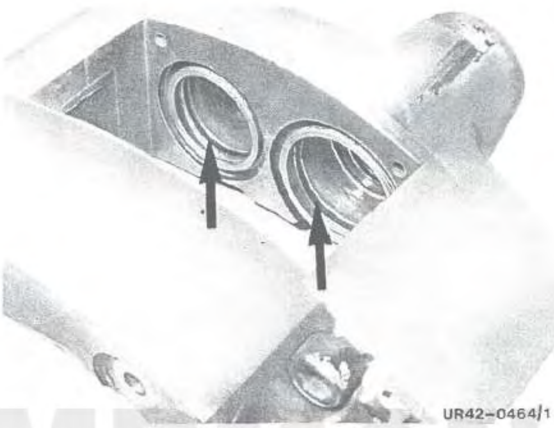


UR42–0464/1

15 Remove sealing rings (24).

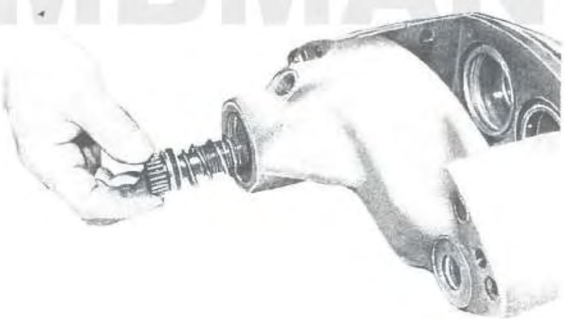
16 Clean all parts with cleaning agent No. 1, check and renew if necessary.

Assembly



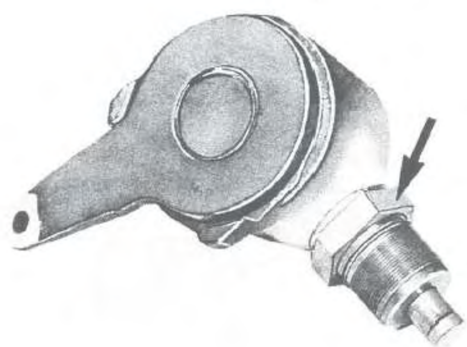
UR42–0464/1

1 Install new sealing rings.



UR42–0463

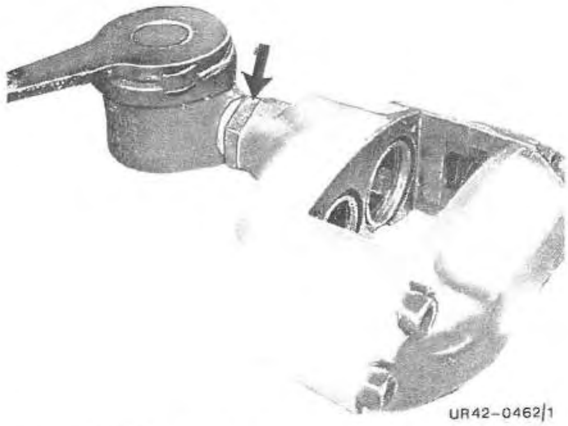
2 Using spray No. 3, spray new drive spindle (19), new sealing ring (21) and new support ring (20). Install drive spindle (19) with compression spring (22).



UR42–0465/1

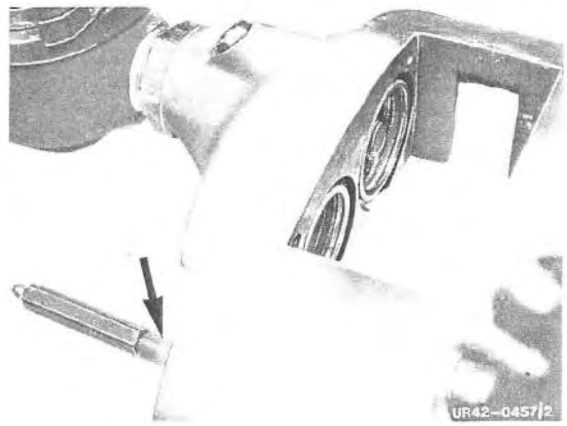
3 Slide new sealing ring (15) on to actuating lever (17). Coat thread and pin with grease No. 2, turn back hexagon nut (16) to stop.

4 Screw actuating lever (17) as far as it will go into the flange section of the housing (23) then slacken lever (max. 1 turn), until it points in direction of bleeder valve.



UR42-0462/1

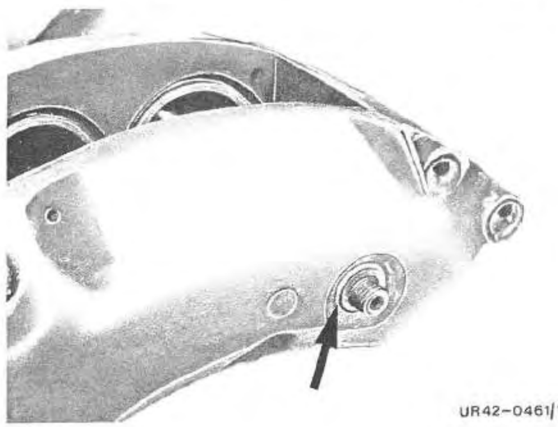
5 Install bleeder valve.
Tightening torque, see 1.3/1.



UR42-0457/2

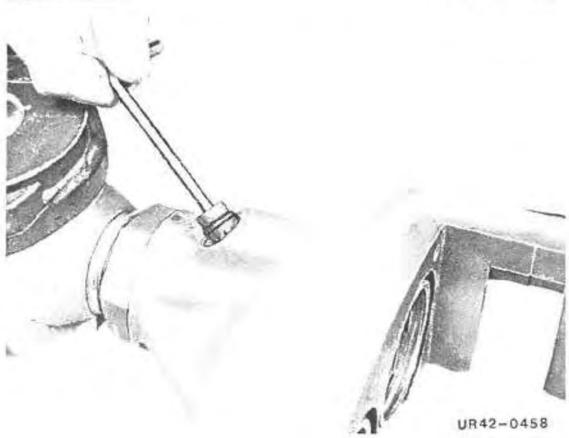
6 Using spray No. 3, spray adjusting spindle (25) and new sealing ring (3) prior to installation.

7 Secure adjusting spindle (25) with lock washer (2).



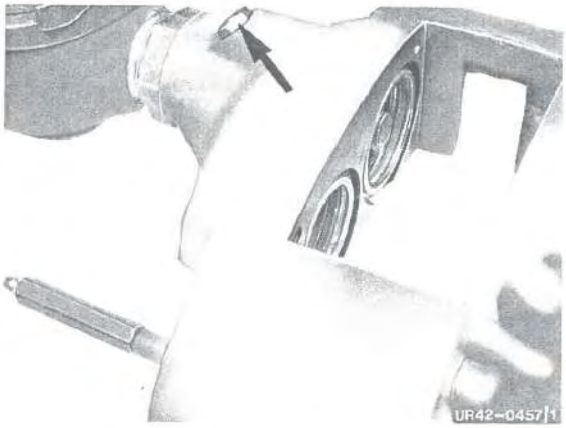
UR42-0461/1

8 Using spray No. 3, spray new restoring shaft (11) and new sealing ring (10) prior to installation.

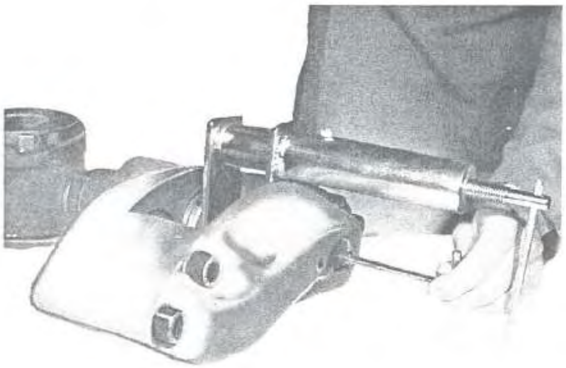


UR42-0458

747.2



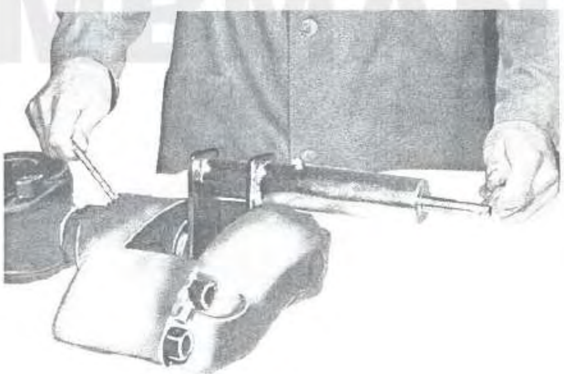
9 Coat threaded sleeve (12) with grease No. 2 and insert with new sealing ring (13).
Tightening torque, see 1.3/1.



UR42-0468

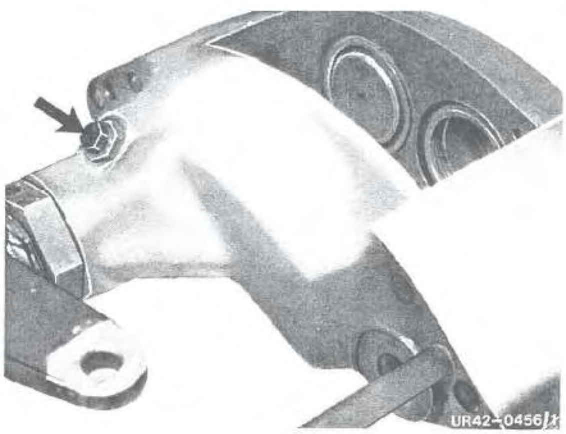
10 Using spray No. 3, spray bore and 57 mm diameter piston. Introduce piston into bore, ensuring the piston is not tilted.

11 Reset piston with adjusting spindle (25) so that piston groove is just left clear. Use special tool No. 11 to apply preload to piston.
Direction of rotation, see 1.5/1.



UR42-0469

12 Install piston as described in steps No. 9 and 10.



13 Tighten screw plug (14) and sealing ring (13).
Tightening torque, see 1.3/1.

14 Using special tool No. 11, fit 48 mm diameter pistons in their bores.

15 Using spray No. 3, spray inside of new protective caps (9), fitting with installation set and special tools No. 11 and No. 14/No. 15.

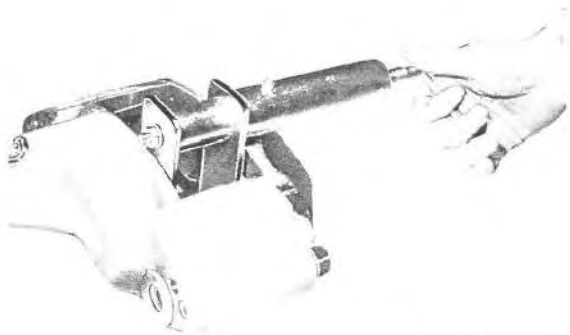
Note: Ensure the pistons are correctly seated.

16 Reset piston until crown is flush with pad shaft.

17 Fit guard caps (8) for pistons using special tools No. 11 and No. 12.

18 Install fixed caliper.

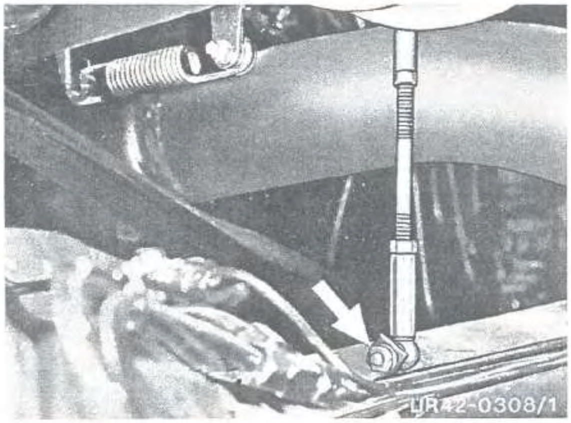
Note: After installation, ensure that actuating lever is correctly located relative to clevis of spring-loaded cylinder; adjust lever if necessary. Tighten hexagon nut to 150 to 180 Nm.



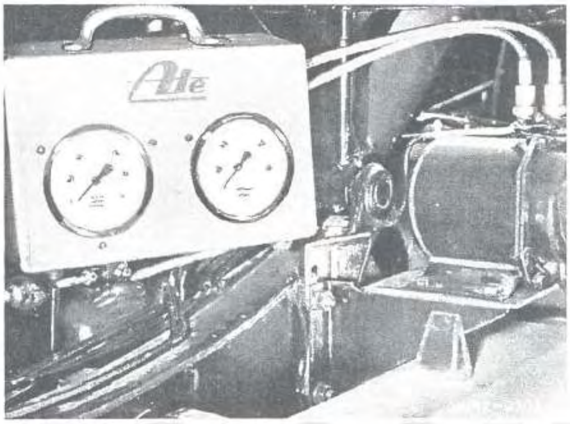
UR42-0467

Checking and Adjusting Failsafe System

1 Detach actuating linkage at torque tube.

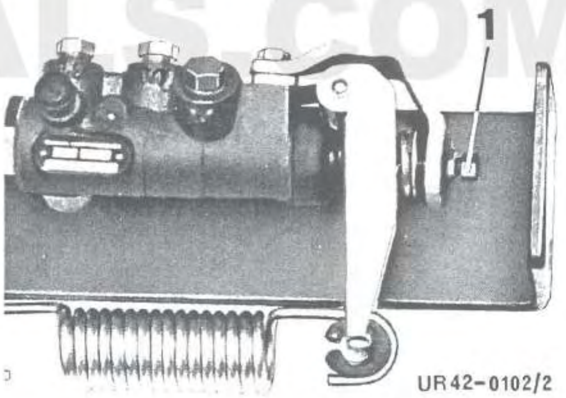


2 Connect a pressure gauge of the tester (special tool No. 3) to each of the two test connections upstream and downstream of the ALB modulator.



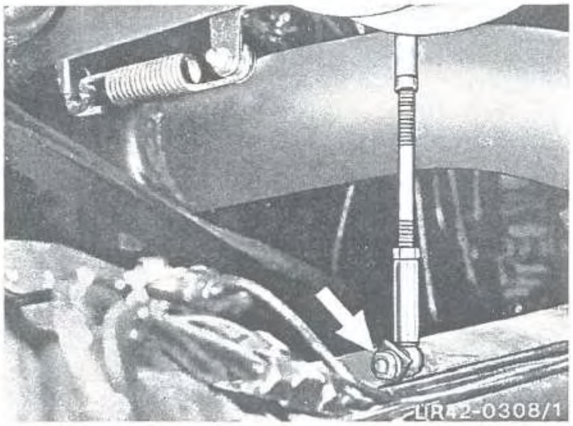
3 Slowly depress the brake pedal. The pressure indicated by both pressure gauges rises uniformily to the failsafe setting of 26 to 30 bar, from this point, the pressure in the modulated circuit must no longer rise.

4 If necessary, adjust the failsafe pressure of 26 to 30 bar using the adjusting screw.



5 Attach actuating linkage to torque tube.

6 Remove both pressure gauges and close off the test connections.



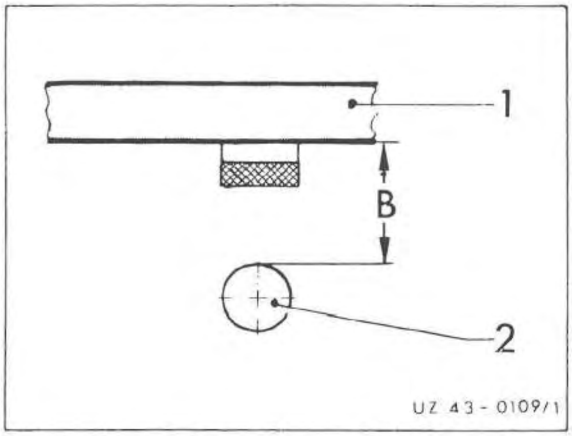
Checking and Adjusting the Load-Dependent Modulation

1 Check failsafe system and adjust if necessary, see 4.1/1.

2 Check modulator linkage for wear and ensure it is correctly secured.

To Chassis End No.
435.110/111 : 086 854
453.113 : 084 641

3 Measure deflection path (distance "B") between lower edge of frame (1) and axle bridge (2).



From Chassis End No.
435.110/111 : 086 855
453.113 : 084 642

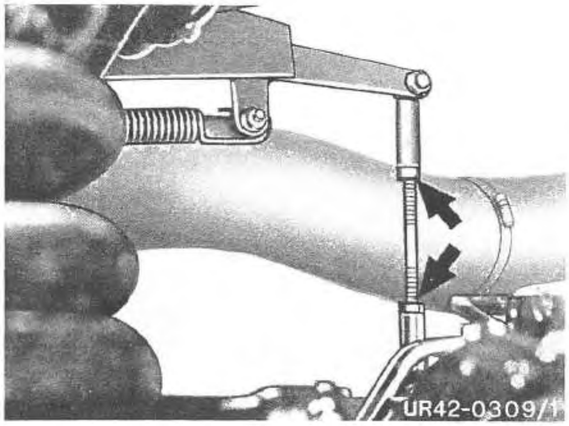
4 Determine rear axle load.

5 Fully depress brake pedal until the brake pressure of the unmodulated circuit reaches approx. 140 bar.
The brake pressure in the modulated circuit must stop within the specified pressure range (see ALB diagram).

6 If necessary, correct modulated pressure by lengthening or shortening the linkage.

Note: Shortening the linkage reduces the pressure. Lengthening the linkage increases the pressure.

7 Unscrew both pressure gauges and close off the test connections.



Directions for Bleeding

The brake system must be bled after any repair job that has involved the opening of the closed hydraulic system, or when the brake pedal feels soft and spongy. A variety of special-purpose fixtures, e. g. the ARC 50 unit or the ATE filler/bleeder, are commercially available for this purpose. Please remember to follow the instructions supplied with these units.

Brake fluid that has been bled off must not be put back because it may contain foreign matter which will then enter the brake system. Besides, brake fluid, being highly hygroscopic, steadily absorbs moisture from the atmosphere and gradually shows a reduced boiling point. As a result, the vapour bubbles may form in the brake system under extreme conditions.

Brake fluid contains constituents which will dissolve paint. Therefore keep well away from vehicle paintwork.

While bleeding is in progress, ensure that bottom of refilling container is permanently covered by brake fluid. On no account must the compensating port be free from fluid because air will otherwise be aspirated when brake pedal is pressed, so rendering the entire operation futile.

Also make sure that the bleed vessel is held high enough to keep level in vessel higher than outlet from bleeder valve.

Bleeding is completed when bubble-free brake fluid runs through the bleeder hose.

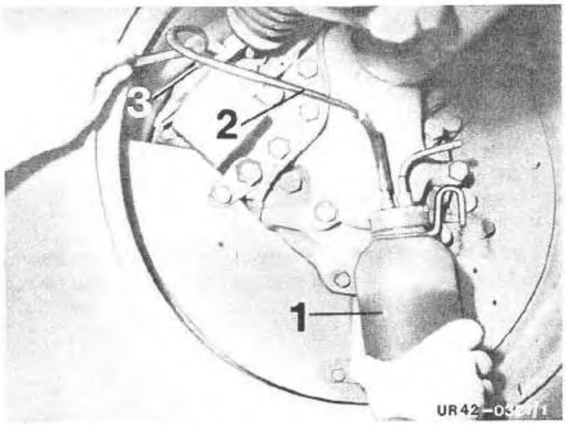
Having completed the entire operation, top up reservoir to mark "maximum" with brake fluid.

Note: If you bleed the system by "pumping" the brake pedal, remember to close the corresponding bleed screw each time before allowing the brake pedal to return. This serves to stop air from being aspirated through the bleed screw hole.

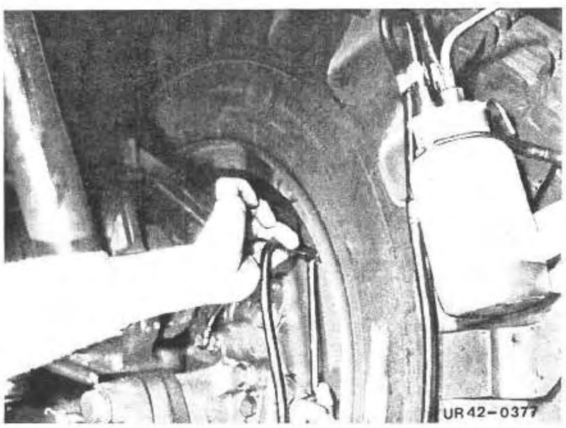
Bleeding

1 Prepare bleeder for use in accordance with manufacturer's specifications.

2 Unscrew caps on both reservoirs and connect special tools No. 5 and No. 6 to the bleeder.



3 Open bleeder screws one after the other on the master brake cylinder, ALB modulator and on the fixed calipers by turning one to two turns and close only when bubble-free brake fluid is obtained at the outlet.



4 Check that the brake system functions correctly.

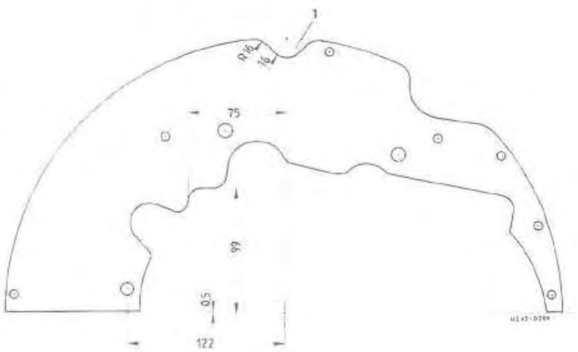
5 Detach bleeder.

Brake Lining Wear Measurement

Note: Special tool No. 2 provides the possibility to check the thickness of the brake lining without removing the wheels provided no cover plates are fitted.

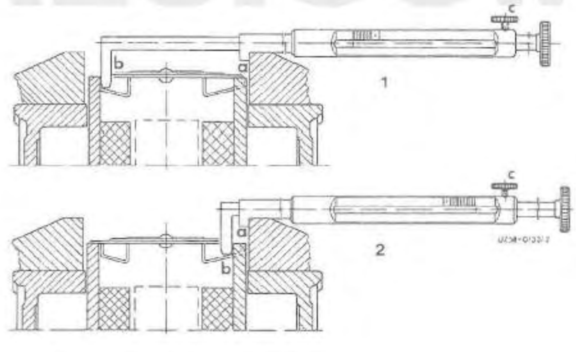
The shaft cover plates should be removed for fixed calipers **without** a brake lining wear indicator.

Mechanical wear measurement cannot be carried out on brake calipers with a brake lining wear indicator without removing the shaft cover plates. It may be possible to determine the thickness from the free calipers.



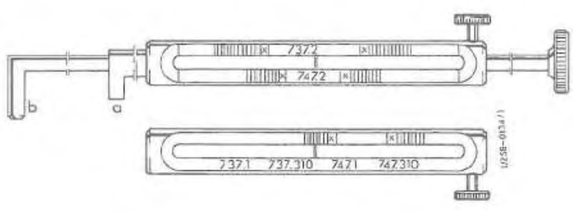
In order to ensure correct use of the special tool, a check must be made as to wether a recess which enables the special tool to be inserted between the wheel rim and the fixed caliper is provided on the brake backplate. If this is not the case, the recess should be provided in the brake backplates (1).

1 Using special tool No. 2 attach the hook "a" to the lower section of the shaft working from the inside of the wheel and place test piece "b" on the brake pad.
Check brake lining thickness on cover side (1).
Check brake lining thickness on flange side (2).



2 Tighten knurled screw "c" and read the brake pad wear off the scale. The brake pad must be renewed if the pointer is in the red area "x" (on the test gauge).

Note: When measuring, ensure the correct scale is used on the test gauge agreeing with the relevant axle model designation.



Contents

Laying:

Plastic pipes must be arranged in such a way to ensure chafing cannot occur and secured with cable clips at 50 cm intervals. It must be possible to still shift the plastic pipe in the tightened cable clip.

When bending the pipe, a minimum radius must be observed otherwise the pipe may kink.

Minimum Bending Diameter for Plastic Pipes

Pipe size Outer diameter x wall thickness	Minimum permissible bending radius (r) in mm
4 x 1	20
6 x 1	30
6 x 0,5	40
8 x 1	40
8 x 2	40
12 x 1,5	60
13 x 1,5	65